

Notes: Li, Lin, and Xu (RFS, 2020)

Political Investment Cycles of State-Owned Enterprises

Contents

1	Big picture	3
2	Data and measurement (sample, ownership, elections, and outcomes)	3
2.1	Election data and political institutions	3
2.2	Ownership data and firm sample	4
2.3	Main outcomes and controls	4
2.4	Matched non-SOE sample	5
3	Models and empirical specifications (all core equations + interpretation)	5
3.1	Baseline investment regression	5
3.2	Matched SOE versus non-SOE comparison	5
3.3	Election timing and close-election interaction	6
3.4	Election outcomes and politicians' incentives	6
3.5	Bank-loan financing equation	6
3.6	Employment-growth equation	7
4	Identification and why the estimates are credible	7
4.1	Baseline identifying idea	7
4.2	What strengthens the causal interpretation	7
4.3	What is still not fully solved	8
5	Tables and figures	8
5.1	Table 1: Summary statistics and sample composition	8
5.2	Table 2: Baseline investment regressions and matched private-firm comparison	8
5.3	Figure 1: Distribution of investment changes around elections	9
5.4	Table 3: Are election results responsive to investment decisions?	9
5.5	Table 4: Robustness tests	10
5.6	Table 5: Fixed timing, flexible timing, and close elections	11
5.7	Table 6: Cross-sectional heterogeneity by institutions and political influence	12

5.8	Table 7: Bank-loan issuance and the financing channel	12
5.9	Internet Appendix evidence on employment creation	13
6	Short summary	13
7	Conclusion	13
7.1	Core conclusion (what the paper establishes)	13
7.2	Economic intuition	14
7.3	Takeaway	14

1 Big picture

- **Question.** Do incumbent politicians use state-owned enterprises (SOEs) to manipulate real economic activity around national elections, and in particular do they push SOEs to invest more in election years to improve reelection odds?
- **Setting.** The paper studies 144,456 SOE firm-year observations around 82 national elections in 25 European countries from 2001 to 2015.
- **Main empirical result.** SOEs increase investment in election years, while matched non-SOEs decrease investment in election years. In the paper’s preferred baseline comparison, SOE investment rises by 0.87 percentage points, which is 28.71% of the SOE sample mean, whereas matched private-firm investment falls by 0.59 percentage points, or 31.12% of the matched private-firm mean.
- **Why that contrast matters.** Private firms behave as the political-uncertainty literature predicts: they postpone investment when elections create uncertainty. SOEs move in the opposite direction, which is exactly what we would expect if they are being used for political objectives rather than pure profit maximization.
- **Mechanism.** Politicians want reelection. Voters respond to investment and employment conditions. SOEs can be pressured to raise investment and hiring, and state-owned banks can be pressured to finance those projects.
- **Where the effect is strongest.** The election-year investment cycle is larger in fixed-timing elections, in closely contested elections, in countries with fewer constraints on executive power, in highly regulated and more centralized political systems, in firms with stronger direct government control, in politically sensitive industries, and in countries with state-dominated banking systems.
- **Financing and labor margins.** SOEs borrow more from banks in election years, especially in countries with state-dominated banking systems, and they also expand employment in election years.
- **Contribution.** The paper gives international firm-level evidence for a political business cycle operating through state-owned firms. It also contributes to the literature on SOE inefficiency by showing one concrete channel: politicians use SOEs to pursue political rather than purely commercial goals.

2 Data and measurement (sample, ownership, elections, and outcomes)

2.1 Election data and political institutions

- **Election sample.** The paper studies 82 national elections in 25 European countries between 2001 and 2015, where the election outcome directly or indirectly determines the national leader.
- **Election sources.** Detailed election information comes mainly from Polity IV and the World Bank Database of Political Institutions, with internet sources used to fill missing details.
- **Why national elections?** They are recurring events with substantial political stakes. They vary across countries and over time, and their timing is external to any individual firm.

- **Institutional-quality measures.** The main cross-country variables are:
 - POLCON: constraints on executive power based on effective veto points.
 - Investment regulation: a reversed Heritage Foundation score, so larger values mean more regulation imposed on business investment.
 - Decentralization: whether subnational governments have residual powers under the constitution.
- **Banking-system measure.** STATE_BANK is the mean share of banking assets held by state-owned banks. This is central for the financing-channel tests.

2.2 Ownership data and firm sample

- **Firm data source.** Ownership and financial data come from Bureau van Dijk’s Amadeus database, which covers both private and public firms across Europe.
- **SOE definition.** A firm is classified as an SOE if its controlling shareholder is a public authority, state, or government, and that owner holds more than 25% of the firm.
- **Period.** The sample begins in 2001 because Amadeus coverage of SOEs becomes comprehensive then and because the authors have annual ownership snapshots from that point onward. The sample ends in 2015.
- **Sample filters.** The authors require total assets above EUR 100,000, drop financial firms and regulated utilities, and exclude countries with fewer than 50 firms during the sample period.
- **Final sample size.** The final sample contains 144,456 firm-year observations.
- **Sample composition.** Russia is large in the sample (about 27% of firm-years), so the paper explicitly checks robustness to dropping Russia and to dropping Russia plus Ukraine.

2.3 Main outcomes and controls

- **Main outcome:** INVESTMENT is yearly growth in fixed assets, scaled by beginning-of-year total assets.
- **Alternative investment measures.**
 - **TOTAL_INVESTMENT:** growth in fixed assets plus growth in inventories, scaled by lagged total assets.
 - **FIX_INVESTMENT:** growth in fixed assets scaled by lagged fixed assets.
- **Main firm controls.** SALE_GROWTH as a proxy for growth opportunities, CASH_FLOW, and in extended specifications size, leverage, cash holdings, profitability, and lagged investment.
- **Main macro controls.** GDP_GROWTH, and in robustness tests growth in government spending, money growth, inflation, and the real interest rate.
- **Financing outcome.** BANKLOAN_ISSUANCE is the change in long-term bank debt plus short-term bank loans, scaled by lagged total assets.
- **Employment outcome.** EMPLOYEE_GROWTH is the change in the log number of employees.

2.4 Matched non-SOE sample

- **Why match?** Average SOEs differ from average private firms, so a raw comparison could just reflect firm characteristics rather than politics.
- **How matching works.** The paper uses one-to-one Mahalanobis-distance matching with replacement, matching SOEs to non-SOEs in the same country and 2-digit NACE industry, using the earliest year in sample and requiring distance below 0.1.
- **Matching variables.** INVESTMENT, CASH_FLOW, size, and SALE_GROWTH.
- **What this buys.** The matched comparison makes the SOE versus non-SOE contrast much more persuasive because the two groups are similar on key observables before the election-year test is run.

3 Models and empirical specifications (all core equations + interpretation)

3.1 Baseline investment regression

The central specification is

$$\text{INVESTMENT}_{i,j,t} = \beta_0 + \beta_1 \text{ELECTION}_{j,t} + \beta_k X_{i,j,t} + \eta_i + \gamma_t + \varepsilon_{i,j,t}. \quad (1)$$

- **Dependent variable.** Firm-level investment by SOE i in country j and year t .
- **Main regressor.** $\text{ELECTION}_{j,t}$ equals one when a national election occurs no earlier than one day after the fiscal year-end in year t and no more than twelve months after fiscal year-end.
- **Controls.** The baseline $X_{i,j,t}$ includes SALE_GROWTH, CASH_FLOW, and GDP_GROWTH.
- **Fixed effects.** Firm and year fixed effects.
- **Inference.** Standard errors are two-way clustered by firm and country $\times$ year.
- **Interpretation of β_1 .** This is the within-firm election-year change in investment after controlling for standard fundamentals and common time shocks. A positive β_1 means SOEs invest more in election years.

3.2 Matched SOE versus non-SOE comparison

To compare SOEs with otherwise similar private firms, the paper estimates a seemingly unrelated system:

$$\text{INVESTMENT}_{i,j,t} = \beta_0 + \beta_1 \text{ELECTION}_{j,t} + \beta_k X_{i,j,t} + \eta_i + \gamma_t + \varepsilon_{1,i,j,t}, \quad (2)$$

$$\text{INVESTMENT}_{n,j,t} = \phi_0 + \phi_1 \text{ELECTION}_{j,t} + \phi_k X_{n,j,t} + \eta_n + \gamma_t + \varepsilon_{2,n,j,t}. \quad (3)$$

- **Interpretation.** The first equation is for SOEs, the second for matched non-SOEs.
- **Why SUR?** It lets the authors estimate both equations jointly and formally test whether the election coefficients differ between SOEs and matched non-SOEs.

- **Key reading.** The sign difference in β_1 versus ϕ_1 is the cleanest reduced-form evidence in the paper that state ownership changes how firms respond to elections.

3.3 Election timing and close-election interaction

To study whether incentives are stronger when election timing is exogenous and outcomes are uncertain, the paper estimates variants of

$$\text{INVESTMENT}_{i,j,t} = \beta_0 + \beta_1 \text{ELECTION}_{j,t} + \beta_2 (\text{ELECTION}_{j,t} \times \text{CLOSE}_{j,t}) + \beta_k X_{i,j,t} + \eta_i + \gamma_t + \varepsilon_{i,j,t}. \quad (4)$$

- **Close-election variable.** CLOSE equals one when the margin of victory is below the first quartile of the cross-country election-margin distribution.
- **Interpretation of β_2 .** A positive β_2 means the SOE investment cycle is stronger precisely when an incumbent has more to gain from swinging the election.
- **Why this matters.** It is a direct political-intent test. If the cycle were purely macroeconomic, it should not get especially stronger in fixed-timing close elections.

3.4 Election outcomes and politicians' incentives

To show why politicians might want to stimulate investment, the paper estimates a country-level probit:

$$\Pr(\text{WIN}_{j,t} = 1) = \Phi(\beta_0 + \beta_1 G_{j,t} + \beta_k X_{j,t}), \quad (5)$$

where $G_{j,t}$ is either GCAP_GROWTH or GFCAP_GROWTH.

- **Dependent variable.** Whether the incumbent government wins the election.
- **Purpose.** This is not the main causal design. It is an incentive-validation exercise showing that higher aggregate investment growth is associated with a higher probability of reelection.
- **Interpretation.** If voters reward aggregate investment, politicians have a reason to pressure SOEs to invest around elections.

3.5 Bank-loan financing equation

The financing-channel regression is

$$\text{BANKLOAN_ISSUANCE}_{i,j,t} = \beta_0 + \beta_1 \text{ELECTION}_{j,t} + \beta_k X_{i,j,t} + \eta_i + \gamma_t + \varepsilon_{i,j,t}. \quad (6)$$

- **Dependent variable.** Net issuance of bank loans scaled by lagged assets.
- **Controls.** Profitability, growth opportunities, tangibility, size, inflation, and GDP growth.
- **Expected sign.** If politically motivated investment needs financing, β_1 should be positive, especially in countries with a large state-owned banking sector.

3.6 Employment-growth equation

The labor-market specification is

$$\text{EMPLOYEE_GROWTH}_{i,j,t} = \beta_0 + \beta_1 \text{ELECTION}_{j,t} + \beta_k X_{i,j,t} + \eta_i + \gamma_t + \varepsilon_{i,j,t}. \quad (7)$$

- **Dependent variable.** Growth in firm employment.
- **Purpose.** To test whether politicians use SOEs not just to raise investment spending but also to support employment before elections.
- **Economic interpretation.** A positive coefficient on ELECTION is exactly what the political-favor view predicts: SOEs create or retain jobs in election years.

4 Identification and why the estimates are credible

4.1 Baseline identifying idea

- **Within-firm election timing.** The paper compares the same SOE across election and nonelection years, with firm fixed effects absorbing time-invariant firm characteristics.
- **Cross-country repetition.** Elections occur in many countries and at different points in time, which helps separate election effects from global or industry-wide trends.
- **Election timing is external to any individual firm.** Even if politicians influence election timing at the country level, any single firm does not choose when the national election occurs.
- **Matched private-firm contrast.** The paper shows that private firms do the opposite of SOEs in election years. That contrast is difficult to square with a simple macro boom story.

4.2 What strengthens the causal interpretation

- **Fixed-timing elections.** The election effect remains in countries where elections are constitutionally fixed, which weakens the concern that incumbents simply call elections when investment is already high.
- **Close-election interaction.** The investment spike is much larger in fixed-timing close elections. This is exactly when the political return to manipulation is highest.
- **Placebo elections.** Randomly generated pseudo-election years do not predict investment.
- **Incumbent not running.** When the incumbent leader is not running for reelection, the election coefficient disappears. That is a powerful political-motivation check.
- **Country $\times$ electoral-cycle fixed effects.** These absorb slow-moving political-cycle factors within each country's government term, so the result is not just comparing one administration to another.
- **Extensive robustness.** The main coefficient survives alternative samples, crisis-year exclusions, extra controls, and dynamic GMM specifications with lagged investment.

4.3 What is still not fully solved

- **No micro-level instrument.** This is strong reduced-form evidence, but it is not an instrument-based design. The paper identifies politically induced cycles through timing, heterogeneity, placebo tests, and contrasts with private firms.
- **Country-year public-policy bundles.** In principle, election years could bring broader public-investment pushes that move SOEs together. The paper treats that as part of the political-use-of-SOEs mechanism, but it is not separately decomposed project by project.
- **Election-outcome regression is supportive rather than definitive.** The probit result that investment growth helps reelection gives incentives, but it is not the main source of identification for the firm-level investment effect.

5 Tables and figures

5.1 Table 1: Summary statistics and sample composition

Read:

- The mean value of INVESTMENT is 0.0303, with substantial dispersion, so there is enough room for economically meaningful election-year movement.
- Election years account for about 24.86% of firm-years.
- Panel B shows the raw unconditional pattern: mean investment is 0.0415 in election years versus 0.0265 in nonelection years, a difference of 0.0150.
- Panel C matters for external validity and robustness. The sample spans 25 European countries, but it is somewhat concentrated, especially in Russia, which is why the paper's leave-Russia-out tests are important.

Table 1					
Descriptive statistics					
A. Summary statistics					
No. of observations	25%	50%	Mean	75%	SD
Firm-year-level variables					
INVESTMENT	144,456	-0.0040	-0.0007	0.0303	0.0335
ELECTION	144,456	0.0000	0.0000	0.2486	0.0000
SALE_GROWTH	144,456	-0.0500	0.0000	0.0519	0.1846
CASH_FLOW	144,456	-0.0002	0.0306	0.0315	0.0865
LEVERAGE	142,617	0.0000	0.0000	0.1101	0.1096
SIZE	144,456	1.2123	2.6307	2.7144	4.1177
ROA	144,322	-0.0205	0.0004	0.0004	0.2546
CASH	138,454	0.0003	0.0401	0.1313	0.1574
BANKLOAN_ISSUANCE	133,210	0.0000	0.0000	0.0330	0.1087
TANGIBILITY	133,210	0.2345	0.5698	0.5282	0.3194
GOVCONTROL	144,456	0.0000	1.0000	0.6542	1.0000
Industry-level variable					
SENSITIVE	652	0.0000	0.0000	0.0736	0.0000
Country-election-level variables					
WV	82	0.0000	0.0000	0.3902	1.0000
GDP_GROWTH	82	-0.0433	0.0275	0.0224	0.1908
COMMON_LAW	82	-0.0164	0.0163	0.0275	0.0720
EDUCATION	82	0.0000	0.0000	0.0610	0.2408
EMPLOYMENT	82	0.0000	0.0000	0.0000	0.0000
CRIME	82	0.0000	0.0000	0.0000	0.0000
DEMOCRACY	82	0.0000	0.0000	0.0000	0.0000
PRESIDENTIAL	82	0.0000	0.0000	0.0000	0.0000
CORRUPTION	82	0.0000	0.0000	0.0000	0.0000
Country-year-level variables					
SPENDING_GROWTH	312	0.0036	0.0142	0.0147	0.0290
GDP_GROWTH	312	0.0044	0.0212	0.0211	0.0357
INFLATION	312	0.0144	0.0242	0.0369	0.0993
STATE_BANK	312	0.0000	0.0000	0.0000	0.0000
LEGAL_GROWTH	146	0.0025	0.1099	0.1469	0.1355
REAL_INTEREST	280	0.0203	0.0410	0.0431	0.0991
B. Mean investment rate (INVESTMENT) of SOEs in election versus nonelection years					
No. of observations	Mean	Median	SD		
Election years	35,905	0.0415	-0.0033	0.2377	
Nonelection years	108,551	0.0265	-0.0051	0.1997	
Difference		0.0150	0.0018		
Test of differences		11.78***	9.61***		

Table 1 (Continued)											
C. Sample composition and mean characteristics by country (25 countries)											
	N	# of elections	Sample period	Election years	Investment	GDP_Growth	POLCON	Investment_Regulation	Decentralization	State_Bank	
Austria	919	3	2006-2015	2006, 2008, 2013	0.0306	0.0131	0.7399	0.2200	Yes	0.0400	
Belgium	1,143	3	2004-2015	2007, 2010, 2014	0.0218	0.0142	0.8922	0.1417	Yes	0.0000	
Bulgaria	2,104	5	2001-2015	2001, 2005, 2009, 2013, 2014	0.0130	0.0161	0.5843	0.4167	No	0.0968	
Croatia	251	2	2002-2012	2003, 2007	0.0238	0.0180	0.7664	0.4778	No	0.1239	
Czech	417	3	2005-2015	2006, 2010, 2013	0.0164	0.0254	0.7559	0.2818	No	0.0843	
Denmark	210	2	2010-2015	2011, 2015	0.0240	0.0019	0.7775	0.1083	No	0.0015	
Finland	1,190	4	2001-2015	2003, 2007, 2011, 2015	0.0285	0.0154	0.7752	0.2507	No	0.0548	
France	7,388	2	2005-2015	2007, 2012	0.0205	0.0110	0.8629	0.4045	No	0.0094	
Germany	11,541	4	2001-2015	2002, 2005, 2009, 2013	0.0173	0.0124	0.8476	0.1207	No	0.3983	
Greece	156	4	2005-2015	2007, 2009, 2012, 2015	0.0320	-0.0091	0.7464	0.4409	No	0.1553	
Ireland	247	2	2003-2015	2007, 2011	-0.0042	0.0255	0.7592	0.0962	No	0.2569	
Italy	9,479	3	2002-2013	2006, 2008, 2013	0.0224	0.0009	0.6847	0.2792	Yes	0.0609	
Latvia	76	3	2005-2015	2006, 2010, 2011	0.0174	0.0415	0.7750	0.2556	No	0.0767	
Luxembourg	76	2	2004-2015	2004, 2009	0.0256	0.0303	0.7657	0.0773	Yes	0.0510	
Netherlands	672	4	2003-2015	2003, 2006, 2010, 2012	0.0232	0.0100	0.7740	0.1000	No	0.0708	
Norway	10,866	4	2001-2015	2001, 2005, 2009, 2013	0.0237	0.0176	0.7714	0.4100	No	0.0000	
Poland	13,246	5	2001-2015	2001, 2005, 2007, 2011, 2015	0.0466	0.0068	0.7352	0.3967	No	0.2738	
Portugal	655	4	2004-2015	2004, 2009, 2011, 2015	0.0197	-0.0011	0.7396	0.3000	No	0.2381	
Romania	2,455	3	2001-2015	2004, 2008, 2012	0.0640	0.0067	0.5948	0.3967	No	0.3990	
Russia	38,620	3	2002-2015	2004, 2008, 2012	0.0209	0.4422	0.6664	0.6786	No	0.4570	
Slovakia	73	3	2005-2015	2006, 2010, 2012	0.0055	0.0419	0.7717	0.2682	No	0.0803	
Spain	5,464	4	2001-2015	2004, 2008, 2011, 2015	0.0212	0.0157	0.8587	0.2500	Yes	0.0000	
Sweden	1,496	2	2003-2015	2006, 2010	0.0208	0.0211	0.7600	0.1364	No	0.0000	
Ukraine	33,125	5	2001-2015	2002, 2006, 2007, 2012, 2014	0.0174	0.0332	0.7284	0.7187	NA	0.1445	
UK	2,587	3	2002-2015	2005, 2010, 2015	0.0076	0.0171	0.7448	0.1429	No	0.0867	
Total	144,456	82		Median	0.0237	0.0171	0.7637	0.2682	No	0.0843	

Panel A reports the summary statistics for the firm-level, industry-level, country-election-level, and country-year-level variables used in the analysis. Panel B reports the summary statistics for investment rates in both election years and nonelection years, where the investment rate (INVESTMENT) is defined as growth in fixed assets scaled by the beginning-of-year book value of total assets. Panel C reports the selected political system and election and country characteristics for each of the twenty-five countries in our sample between 2001 and 2015. The number of observations indicates the total number of firm-year observations included in Bureau van Dijk's Amadeus database for each country in our sample. A country is classified as having flexible timing if the national leader or legislative body has the option to call an election before the regularly scheduled election date. See Table A1 for variable descriptions and sources. *** $p < .01$.

5.2 Table 2: Baseline investment regressions and matched private-firm comparison

Read:

- Panel A shows the matching works: SOEs and matched non-SOEs look very similar on cash flow, size, sales growth, and investment in the pre-match year.
- Panel B is the paper’s headline result. In the fully controlled SUR specification, the election coefficient is +0.0087 for SOEs and −0.0059 for matched non-SOEs.
- Economically, SOEs increase investment by 28.71% of their sample mean, while matched private firms cut investment by 31.12% of their sample mean.
- The difference between the election coefficients is highly significant. This is the clearest evidence that state ownership changes firms’ response to political cycles.

Table 2
Baseline investment regressions

A. Comparing SOEs and non-SOEs in the matched sample

	25th percentile		Median		75th percentile		p-value of test of equality of medians	p-value of test of equality of distributions
	SOEs	Matched non-SOEs	SOEs	Matched non-SOEs	SOEs	Matched non-SOEs		
Matching variables								
CASH_FLOW	0.0026	0.0030	0.0327	0.0338	0.0842	0.0844	.2888	.4930
SIZE	0.0666	0.0256	2.3821	2.8601	3.5577	3.5359	.3491	.7980
SALE_GROWTH	−0.0622	−0.0608	0.0476	0.0433	0.1831	0.1796	.7205	.7210
INVESTMENT expenditures	−0.0280	−0.0272	−0.0052	−0.0049	0.0082	0.0081	.4658	.4630

B. Seemingly unrelated regressions for SOE and matched non-SOE subsamples

Dependent variable = INVESTMENT	SUR(1)		SUR(2)		SUR(3)		SUR(4)	
	1A	1B	2A	2B	3A	3B	4A	4B
	SOEs	Matched non-SOEs	SOEs	Matched non-SOEs	SOEs	Matched non-SOEs	SOEs	Matched non-SOEs
ELECTION	0.0090*** (2.71)	−0.0050*** (−2.16)	0.0089*** (2.69)	−0.0052*** (−2.23)	0.0089*** (2.70)	−0.0051*** (−2.41)	0.0087*** (2.69)	−0.0059*** (−2.66)
SALE_GROWTH			0.0075*** (3.53)	0.0075*** (3.53)	0.0075*** (3.53)	0.0075*** (3.53)	0.0075*** (3.53)	0.0070*** (3.54)
CASH_FLOW			0.0944*** (7.36)	0.1278*** (14.03)	0.0944*** (7.35)	0.1278*** (14.03)	0.0944*** (7.35)	0.1268*** (13.94)
GDP_GROWTH					0.1206*** (1.79)	0.1617*** (0.09)		
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Test of differences of ELECTION coefficients between SOE subsample and matched non-SOE subsample

p-value	.000***	.000***	.000***	.000***
No. of observations	261,392	261,392	261,392	261,392
Adjusted R ²	.071	.072	.078	.078

Panel A compares the summary of matching variables between SOEs and non-SOE firms in the matched sample. Panel B reports the results for SOE subsample and matched non-SOE subsample using seemingly unrelated regressions (SUR) based on the systems of equations (2A) and (2B):

$$INVESTMENT_{i,t} = \beta_0 + \beta_1 ELECTION_{i,t} + \beta_2 SIZE_{i,t} + \beta_3 CASH_FLOW_{i,t} + \beta_4 GDP_GROWTH_{i,t} + \eta_i + \eta_t \quad (2A)$$

$$INVESTMENT_{i,t} = \alpha_0 + \alpha_1 ELECTION_{i,t} + \alpha_2 SIZE_{i,t} + \alpha_3 CASH_FLOW_{i,t} + \alpha_4 GDP_GROWTH_{i,t} + \eta_i + \eta_t \quad (2B)$$

ELECTION is an election year indicator variable that equals one for any firm-year in which an election takes place. To create the matched non-SOE sample, we adopt the Mahalanobis distance matching approach. The distances between SOEs and non-SOE firms are computed from a set of variables including INVESTMENT, CASH_FLOW, SIZE, and SALE_GROWTH. We require one-to-one matching with replacement, and firms in the treated group and control group are from the same 2-digit NACE industry and the same country. The earliest year in the sample period is used in constructing the match. Additionally, we also require that Mahalanobis distance between SOEs and non-SOE firms is less than 0.1. Intercept, year and firm fixed effects are included in all regressions but are not reported. Asterisks denote statistical significance using standard errors two-way clustered at the firm and country-year levels and the associated

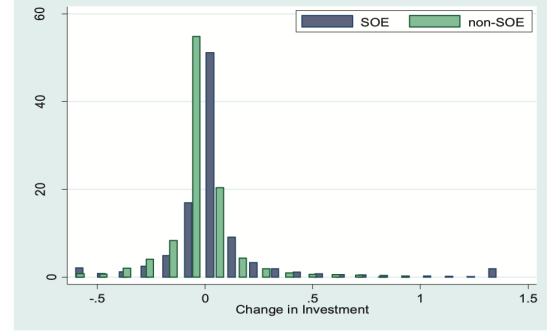


Figure 1
Distribution of investment changes for SOEs and non-SOEs around elections

The figure shows the distribution of investment changes for the SOEs and the matched non-SOEs in the regression sample. Investment changes are calculated as the change in investment from the year before the election year to the election year, scaled by the investment in the year before the election year. To create the matched non-SOE subsample, we adopt the Mahalanobis distance matching approach. The distances between SOEs and non-SOE firms are computed from a set of variables including INVESTMENT, CASH_FLOW, SIZE, and SALE_GROWTH. We require one-to-one matching with replacement, and firms in the treated group and controls group are from the same 2-digit NACE industry and the same country, and the earliest year in the sample period is used in constructing the match. Additionally, we also require that the Mahalanobis distance between SOEs and non-SOE firms be less than 0.1.

5.3 Figure 1: Distribution of investment changes around elections

Read:

- The figure is a visual version of Table 2. The SOE distribution shifts to the right, while the matched non-SOE distribution shifts to the left.
- The average change in investment around elections is +0.0359 for SOEs and −0.0090 for matched non-SOEs.
- This figure is useful because it shows the baseline contrast is not being driven by only a few extreme observations.

5.4 Table 3: Are election results responsive to investment decisions?

Read:

- Higher aggregate investment growth predicts a higher probability that the incumbent wins. The marginal effect is 24.62% for gross capital formation growth and 22.48% for gross fixed capital formation growth when moving from the first to the third quartile.
- This is not the main causal result of the paper. Instead, it is the incentive link: politicians have electoral reasons to care about investment conditions.

Table 3
Are election results responsive to investment decisions?

Dependent variable = WIN	(1)	(2)	(3)	(4)
	Coef. (z-stat.)	Marginal Δ in prob. (%)	Coef. (z-stat.)	Marginal Δ in prob. (%)
<i>GCAP_GROWTH</i>	5.1200** (2.49)	24.62%		
<i>GFCAP_GROWTH</i>			7.8431*** (2.86)	22.48
<i>COMMON_LAW</i>	1.1882** (2.42)	44.62%	1.2501** (2.53)	46.68
<i>EDUCATION</i>	2.2713 (1.63)	15.72%	2.2772* (1.70)	15.56
<i>INFLATION</i>	-3.2059 (-0.29)	-2.33%	-1.4520 (-0.13)	-1.04
<i>EMPLOYMENT</i>	2.4824 (0.28)	4.13%	0.8150 (0.09)	1.33
<i>CRIME</i>	-30.4650 (-1.07)	-12.03%	-31.7292 (-1.05)	-11.44
<i>DEMOCRACY</i>	-12.6813* (-1.93)	-42.49%	-12.3948* (-1.79)	-41.13
<i>PRESIDENTIAL</i>	0.3252 (0.42)	11.56%	0.3988 (0.53)	14.20
<i>CORRUPTION</i>	-6.8454** (-2.33)	-76.93%	-7.3848** (-2.42)	-80.21
Year fixed effects	Yes		Yes	
No. of elections	82		82	
Pseudo R^2	.344		.362	

This table shows the results of the probit regression based on Equation (3):

$$Prob(WIN)_{j,t} = \beta_0 + \beta_1 G_{j,t} + \beta_k X_{j,t} + \epsilon_{j,t}. \quad (3)$$

The dependent variable, $WIN_{j,t}$, is an indicator variable that equals one if the incumbent government wins the election and zero otherwise. The independent variable, $G_{j,t}$, refers to gross capital formation growth (*GCAP_GROWTH*), or gross fixed capital formation growth (*GFCAP_GROWTH*). *GCAP_GROWTH* is measured as the percentage change in gross capital formation (formerly gross domestic investment) over the previous year, and *GFCAP_GROWTH* is measured as the percentage change in gross fixed capital formation (formerly gross domestic fixed investment) over the previous year. Year fixed effects are included. A marginal change in probability is calculated as the change in the likelihood of winning an election when the underlying variable changes from the first to the third quartile of the sample distribution or from 0 to 1 in the case of dummy variables, holding all other independent variables at their respective means. Asterisks denote statistical significance using standard errors two-way clustered at the country and year levels, and the associated z-statistic is reported in parentheses below the coefficients. Table A1 defines the variables. * $p < .1$; ** $p < .05$; *** $p < .01$.

- The paper also notes that SOE investment is a meaningful share of total investment, so pushing SOEs to invest can move an aggregate variable that voters observe.

5.5 Table 4: Robustness tests

Table 4
Baseline investment regressions: Robustness tests

A. Alternative samples						
Dependent variable = INVESTMENT	(1)	(2)	(3)	(4)	(5)	(6)
Placebo test	Country-industry-year medians	Crisis country-years omitted	Incumbent not running	Finland, France, and Poland omitted	Russia excluded	Ukraine and Russia excluded
<i>ELECTION</i>	-0.0004 (-0.09)	0.0054* (1.73)	0.0157*** (3.81)	-0.0055 (-0.79)	0.0104*** (2.58)	0.0088** (2.55)
Firm-level controls	Yes	Yes	Yes	Yes	Yes	Yes
Macro-level controls	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	No	Yes	Yes	Yes	Yes
Industry fixed effects	No	Yes	No	No	No	No
Country fixed effects	No	Yes	No	No	No	No
No. of observations	144,456	9,759	52,680	144,456	122,632	105,836
Adjusted R^2	.061	.033	.099	.061	.075	.090
B. Control for country $\times$ electoral cycle, firm and year fixed effects						
Dependent variable = INVESTMENT	(1)	(2)	(3)	(4)	(5)	(6)
<i>ELECTION</i>	0.0095*** (2.96)	0.0093*** (2.91)	0.0094*** (2.96)	0.0094*** (2.96)	0.0090*** (2.89)	0.0086*** (2.86)
<i>SALE_GROWTH</i>		0.0081*** (3.73)	0.0075*** (3.48)	0.0075*** (3.48)	0.0072*** (3.36)	0.0072*** (3.36)
<i>CASH_FLOW</i>			0.0062*** (7.35)	0.0062*** (7.35)	0.0062*** (7.35)	0.0062*** (7.35)
<i>GDP_GROWTH</i>						0.1529*** (2.36)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Country $\times$ Electoral cycle fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
No. of observations	144,456	144,456	144,456	144,456	144,456	144,456
Adjusted R^2	.057	.057	.061	.061	.061	.061

Table 4
(Continued)

C. Alternative model specifications						
Dependent variable = INVESTMENT	(1)	(2)	(3)	(4)	(5)	(6)
Controls included	Extra micro-controls Size, leverage cash, age and profitability	Growth in central spending	M2 growth	Real interest rate	Inflation	Lagged investment
<i>ELECTION</i>	0.0079*** (2.30)	0.0084*** (2.51)	0.0099*** (2.35)	0.0082*** (3.35)	0.0097*** (3.35)	0.0133*** (3.96)
<i>INVESTMENT(-1)</i>						0.0597*** (8.64)
Firm-level controls	Yes	Yes	Yes	Yes	Yes	Yes
Macro-level controls	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
No. of observations	156,516	144,456	105,424	137,835	144,456	108,228
Adjusted R^2	.132	.061	.062	.061	.061	.061

This table reports additional tests and robustness checks. Panel A, Column 1, presents the estimation results for the placebo test. To conduct the placebo test, for each country, we randomly select a starting date within the first 5 years of the sample that is not already an election year. The randomly selected date becomes the new election date, and we construct the remaining election dates based on the average periodicity of elections within each country. We then define *ELECTION* as before, but with reference to the randomly selected date. We then report our main regressions in Table 2. Panel A, Column 2, presents the estimation results using country-industry-year medians; Column 3 reports the estimation results excluding countries that experienced a financial crisis during the sample period based on a study by Laeven and Valencia (2012); Column 4 reports the estimation results when considering only those elections in which the incumbent is not running for reelection; Column 5 presents the results when Finland, France and Poland are omitted; Column 6 presents the results when Russia is excluded; and Column 7 presents the results when Ukraine and Russia are excluded. Panel B, Columns 1 to 4, report the baseline investment regression results for the effect of national elections on SOE investment with country $\times$ electoral cycle and year fixed effects. Panel C, Column 1, reports the specification that includes macro-level controls, such as firm size (*SIZE*), leverage (*LEVERAGE*), cash holdings (*CASH*), and profitability (*ROA*). Columns 2, 3, 4, and 5 report the specification that includes other macro-level controls such as the growth in central government spending over the previous year (*SPENDING_GROWTH*), the growth in the money supply (*M2_GROWTH*), the real interest rate (*REAL_INTEREST*), and the inflation rate (*INFLATION*), respectively. Column 6 (7) reports the estimation results controlling for the lagged *INVESTMENT* based on the one-step (two-step) Arellano and Bond (1991) GMM estimation. Intercept and fixed effects are included in all regressions but are not reported. Asterisks denote statistical significance using standard errors two-way clustered at the firm and country-year levels, but statistical significance is used standard errors two-way clustered at the industry and country $\times$ year levels for Column 2 in panel A, and associated *t*-statistics are reported in parentheses below the coefficients. Table A1 defines the variables. * $p < .1$; ** $p < .05$; *** $p < .01$.

Read:

- The placebo election dummy is insignificant, which is exactly what we want.
- The effect survives country-industry-year medians, crisis-year exclusions, dropping Russia, and dropping Russia plus Ukraine.
- When the incumbent leader is not running for reelection, the election coefficient becomes insignificant. That result is especially persuasive because it ties the effect directly to incumbents' private political incentives.

- With country×electoral-cycle fixed effects, the coefficient remains around 0.0080; with lagged investment in GMM, it remains around 0.0079 to 0.0133.

5.6 Table 5: Fixed timing, flexible timing, and close elections

Read:

- The election coefficient is positive in both fixed- and flexible-timing subsamples, so the main result is not driven by endogenous election timing alone.
- The key extra result is the interaction with close elections. In fixed-timing elections, the ELECTION ×CLOSE coefficient is 0.0228, large and significant.
- The paper translates this into a 91.75% increase in investment relative to the sample mean for close fixed-timing elections, compared with a much smaller 16.50% increase for nonclose fixed-timing elections.
- In flexible-timing elections, the close-election interaction is not significant, which is consistent with incumbents timing elections when conditions already favor them.

Table 5
Elections and investment: Election timing and predictability of outcomes

Dependent variable = INVESTMENT	(1)	(2)	(3)	(4)
	Fixed election timing		Flexible election timing	
ELECTION	0.0059** (2.14)	0.0050* (1.71)	0.0084** (2.18)	0.0080** (1.97)
ELECTION×CLOSE		0.0228** (2.21)		0.0033 (0.56)
SALE_GROWTH	0.0106*** (2.80)	0.0106*** (2.80)	0.0040* (1.81)	0.0039* (1.81)
CASH_FLOW	0.0522*** (4.50)	0.0522*** (4.50)	0.1519*** (10.74)	0.1519*** (10.74)
GDP_GROWTH	0.0959 (1.57)	0.0936 (1.56)	0.137*** (2.74)	0.140*** (2.85)
Year fixed effects	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
No. of observations	61,152	61,152	83,304	83,304
No. of countries	7	7	18	18
No. of elections	18	18	64	64
Adjusted R ²	.048	.048	.075	.075

This table presents the estimation results for election timing and predictability of outcomes. *CLOSE* is an indicator variable that equals one if the difference in votes between the winner and the runner-up of the election is below the 25th percentile of the victory margin distribution over the sample of countries under consideration. Columns 1 and 2 present the results for countries with fixed election timing; Columns 3 and 4 present the results for countries with flexible election timing. A country is classified as having flexible timing if the national leader or legislative body has the option to call an election before the regularly scheduled election date. Intercept, year fixed effects, and firm fixed effects are included in all regressions but are not reported. Asterisks denote statistical significance using standard errors two-way clustered at the firm and country×year levels, and the associated *t*-statistics are reported in parentheses below the coefficients. Table A1 defines the variables. **p* < .1; ***p* < .05; ****p* < .01.

Table 6
Elections and investment: Institutional quality and political influence characteristics

A. Cross-sectional tests: Institutional quality						
Dependent Variable= INVESTMENT	(1)	(2)	(3)	(4)	(5)	(6)
	Political constraint		Investment regulation		Decentralization	
	High	Low	High	Low	Yes	No
ELECTION	0.0021 (0.87)	0.0093** (2.31)	0.0096*** (2.61)	0.0015 (0.49)	0.0049 (1.23)	0.0089*** (3.57)
SALE_GROWTH	0.0051** (2.19)	0.0078*** (2.80)	0.0080*** (3.12)	0.0036 (1.38)	−0.0008 (−0.30)	0.0093*** (3.04)
CASH_FLOW	0.0553*** (2.99)	0.1021*** (6.55)	0.0931*** (6.71)	0.1005*** (3.95)	0.139*** (4.58)	0.0624*** (5.79)
GDP_GROWTH	0.2254*** (3.27)	0.0655 (0.90)	0.1015 (1.49)	0.2701*** (3.50)	0.2095 (1.53)	0.1090** (2.01)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Test of differences of ELECTION coefficients between two groups						
p-value		.027**		.043**		.543
No. of observations	40,446	104,010	118,762	25,694	17,081	94,250
Adjusted R ²	.083	.059	.059	.137	.094	.065
B. Cross-sectional tests: Political influencing characteristics						
Dependent variable = INVESTMENT	(1)	(2)	(3)	(4)	(5)	(6)
	Absolute government controlled SOE		Political sensitive industry		State-dominated banking system country	
	Yes	No	Yes	No	Yes	No
ELECTION	0.0140*** (3.95)	0.0075** (2.57)	0.0199*** (3.41)	0.0063* (1.84)	0.0097** (2.43)	0.0028 (1.07)
SALE_GROWTH	0.0066** (2.00)	0.0037 (1.28)	0.0107* (1.76)	0.0068*** (3.30)	0.0077*** (2.86)	0.0056** (2.32)
CASH_FLOW	0.0904*** (5.47)	0.0872*** (5.39)	0.1219*** (4.98)	0.0891*** (6.31)	0.105*** (6.73)	0.0410** (2.32)
GDP_GROWTH	0.1841*** (3.61)	0.0316 (0.50)	−0.0025 (−0.03)	0.1357** (1.97)	0.0992 (1.38)	0.2905** (2.58)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Test of differences of ELECTION coefficients between two groups						
p-value		.068*		.002***		.042**
No. of observations	94,504	49,952	24,429	120,027	114,466	29,990
Adjusted R ²	.050	.142	.054	.063	.059	.072

This table reports the Equation (1) regression results based on institutional quality and political influence characteristics. Columns 1 to 2 in Panel A report the results for the association between SOE investment and national elections, conditional on the political constraint index (*POLCON*). A low *POLCON* value indicates high electoral uncertainty because fewer political constituencies are needed to make large policy changes. We partition countries based on whether *POLCON* is above the median level into HIGH versus LOW groups. Columns 3 to 4 report the regression results conditional on government regulations on investment. *INVESTMENT_REGULATION* is a measure of the regulations imposed on investment by a government and a higher value indicates more regulations imposed. We partition countries into HIGH versus LOW groups based on whether *INVESTMENT_REGULATION* is above the median level. Columns 5 to 6 report the results of the association conditional on country decentralization (*DECENTRALIZATION*). A country is considered decentralized if subnational governments have residual powers under the constitution, and zero otherwise. Columns 1 and 2 in panel B report the results for the association between SOE investment and national elections, conditional on whether a government has 50% or more ownership in an SOE. Columns 3 to 4 in panel B report the results for the association between SOE investment and national elections, conditional on whether an SOE belongs to a politically sensitive industry. Columns 5 to 6 in panel B report the results for the association between SOE investment and national elections, conditional on the level of state-owned banks in a country. A bank is considered state owned (*STATE_BANK*) if 50% or more of its shares are controlled by the government. We partition countries into countries with state- and non-state-dominated banking systems based on whether *STATE_BANK* is above the median level of sample countries. Intercept, year fixed effects, and firm fixed effects are included in all regressions but are not reported. Asterisks denote statistical significance using standard errors two-way clustered at the firm and country×year levels, and associated *t*-statistics are reported in parentheses below the coefficients. Table A1 defines the variables. **p* < .1; ***p* < .05; ****p* < .01.

Table 7
Elections and SOEs' bank loan issuance: The role of state-owned banks in financing

Dependent variable= <i>BANKLOAN_ISSUANCE</i>	(1)	(2)	(3)
	Full sample	State-dominated banking system country SYSTEM	
		Yes	No
<i>ELECTION</i>	0.0024** (2.15)	0.0049*** (3.77)	0.0001 (0.03)
<i>TANGIBILITY</i>	-0.0013 (-0.33)	0.0030 (0.78)	-0.0140 (-1.04)
<i>SIZE</i>	-0.0110*** (-6.40)	-0.0095*** (-5.62)	-0.0235*** (-5.76)
<i>ROA</i>	-0.0095*** (-4.40)	-0.0096*** (-4.63)	-0.0062 (-0.68)
<i>SALE_GROWTH</i>	0.0010 (1.54)	0.0006 (1.02)	0.0031 (1.39)
<i>INFLATION</i>	-0.0514*** (-3.30)	-0.0195 (-1.06)	-0.0957 (-0.74)
<i>GDP_GROWTH</i>	-0.0275 (-1.49)	0.0027 (0.16)	0.0019 (0.02)
Year fixed effects	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Test of differences of <i>ELECTION</i> coefficients between two groups			
<i>p</i> -value		.072*	
No. of observations	133,210	106,676	26,534
Adjusted <i>R</i> ²	.433	.496	.288

This table reports the regression results for the effect of national elections on SOEs' bank loan issuance based on Equation (4):

$$BANKLOAN_ISSUANCE_{i,j,t} = \beta_0 + \beta_1 ELECTION_{j,t} + \beta_k X_{i,j,t} + \varepsilon.$$

BANKLOAN_ISSUANCE represents SOEs' bank loan issuance and is measured as the change in long-term bank debt plus short-term bank loans in year *t* scaled by the total assets at the beginning of the year. Column 1 reports the regression results for the association between bank loan issuance and national elections, based on the full sample. Columns 2 and 3 report the regression results for the association between bank loan issuance and national elections, conditional on whether the SOEs reside in countries with state-dominated banking systems. *STATE_BANK* captures the level of state-owned banks in a country, and is measured as the mean proportion of banking assets in state-owned banks between 1999 and 2011. A bank is considered state owned if the government controls 50% or more of its shares. We partition the countries based on whether *STATE_BANK* is above the median level into state- and non-state-dominated banking system countries. Intercept, year fixed effects, and firm fixed effects are included in all regressions but are not reported. Asterisks denote statistical significance using standard errors two-way clustered at the firm and country×year levels, and the associated *t*-statistics are reported in parentheses below the coefficients. Table A1 defines the variables. **p* < .1; ***p* < .05; ****p* < .01.

5.7 Table 6: Cross-sectional heterogeneity by institutions and political influence

Read:

- **Institutional quality.** The election effect is much stronger in low-POLCON countries and in highly regulated countries. The decentralization split goes in the predicted direction too, though the difference is not statistically sharp.
- **Direct political control.** SOEs with absolute government control (government ownership above 50%) have a larger election-year investment spike than other SOEs.
- **Industry sensitivity.** The effect is strongest in politically sensitive industries such as health care, defense, petroleum, telecommunications, and transportation.
- **State-owned banks.** The election effect is stronger in countries with more state-owned banks, which already hints at the financing channel developed in Table 7.

5.8 Table 7: Bank-loan issuance and the financing channel

Read:

- In the full sample, SOEs increase bank borrowing in election years by 0.0024, which is 7.27% of the sample mean bank-loan issuance.
- The result is much stronger in countries with state-dominated banking systems: the coefficient is 0.0049, or 14.85% of the sample mean.
- In countries without state-dominated banking systems, the coefficient is essentially zero.

- This is the core financing-channel evidence. Politicians appear to use state-owned banks together with SOEs.

5.9 Internet Appendix evidence on employment creation

Read:

- The paper's employment regressions show that SOEs also expand employment in election years.
- The estimated increase in employee growth is 1.69% in the full sample and about 1.95% in countries with state-dominated banking systems.
- Additional appendix evidence shows stronger investment responses in labor-intensive industries, which fits the view that politicians value job creation as part of election strategy.

6 Short summary

- **Conclusion.** The paper shows that SOEs invest more in national election years, unlike comparable private firms, and that this pattern is strongest when political incentives and political control are strongest.
- **Intuition.** Politicians want reelection; voters reward economic activity and employment; SOEs and state-owned banks can be used to push both upward at election time.
- **Empirical lesson.** Firm ownership structure fundamentally changes how firms respond to political uncertainty. Private firms delay investment; SOEs can be mobilized to offset that delay for political purposes.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** SOEs exhibit a politically induced investment cycle around national elections.
- **Magnitude.** The baseline election-year increase in SOE investment is about 28.71% of the sample mean; matched non-SOEs move in the opposite direction.
- **Political mechanism.** The effect is stronger when incumbents face tighter electoral competition, stronger control rights, fewer institutional constraints, and greater access to state-owned bank finance.
- **Financing evidence.** SOEs borrow more from banks in election years, especially where the banking system is state dominated.
- **Labor-market evidence.** SOEs also increase employment growth in election years.
- **Broader interpretation.** These patterns are hard to reconcile with pure profit maximization and instead line up with the view that SOEs have dissonant objectives and are used to pursue political goals.

7.2 Economic intuition

- **Why politicians use SOEs.** Politicians need instruments that they can influence quickly and selectively. SOEs are ideal because the state already has control rights.
- **Why investment is attractive.** New projects are visible, can raise measured aggregate investment, and can create jobs or at least the appearance of economic momentum before voting.
- **Why private firms do not behave the same way.** Private firms care about profits and often treat elections as uncertainty shocks, so they wait. That is why the SOE-private comparison is so telling.
- **Why state-owned banks matter.** Politically convenient projects may not be privately profitable, so financing frictions would matter unless the state also influences lenders.

7.3 Takeaway

This is a strong and intuitive political-economy paper. Its main message is that elections do not affect all firms in the same way: private firms often pull back, but SOEs can be pushed forward by politicians seeking reelection. The paper is especially convincing because the real-side investment result, the bank-financing channel, the employment evidence, and the close-election heterogeneity all line up around the same political mechanism.